

Othmen Bourbia

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PROFILE INFO

I am a third-year Industrial Engineering student at ENIT, passionate about research in supply chain, AI, and data science. I like working in labs to build innovative models and optimize industrial processes for Industry 4.0. I am looking for a research project for my end-of-studies internship. My goal is to continue my academic path with a Master's degree and a PhD.

AREAS OF EXPERTISE & TECHNICAL SKILLS

● Mathematics & Statistics Linear algebra, Markov chains, Mixed Integer Linear Programming, Heuristics, Operations research, Stochastic processes.	● Machine Learning & AI Predictive Modeling, Reinforcement Learning, TensorFlow, Scikit-Learn, Deep Learning.	● Optimization & OR SIMPLEX, PuLP, Excel Solver, OR-Tools, CPLEX, Gurobi.
● Industrial Systems & Sim. Arena, Digital Twins, Predictive Maintenance, Markov Chains.	● Data & Analytics SQL, Power BI, Preprocessing, Statistical Modeling, Excel.	● Programming & Tools Python, C, C++, R, Git, Docker, LaTeX, SolidWorks.

EDUCATION

National Engineering School of Tunis (ENIT) <i>Industrial Engineering Degree</i>	<i>September 2024 - Present</i> <i>Tunis, Tunisia</i>
Preparatory Institute for Engineering Studies of El Manar <i>Pre-University education, specializing in Physics/technics</i>	<i>September 2022 - July 2024</i> <i>Tunis, Tunisia</i>
Bab El Khadra High school <i>High School education, specializing in technics— Laureate of Promotion</i>	<i>June 2022</i> <i>Tunis, Tunisia</i>

ACADEMIC RESEARCH AND INDUSTRIAL EXPERIENCE

Strategic and Preventive Lifecycle Tracking System for Cutting Tools <i>Engineering Internship</i>	<i>Figeac Aero, Tunisia</i> <i>June 2026 – August 2026</i>
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- Built a predictive maintenance app to track and classify cutting-tool wear into 5 health stages based on machining session parameters (speed, duration, and tool type).
- Applied Markov Chains to model wear patterns and predict tool degradation over the next 1 to 3 uses.
- Developed a web dashboard with secure login, analytics, alerts, and CSV import to guide maintenance decisions.

Modeling and Simulation of a Fitness Center <i>Simulation Research Project</i>	<i>ENIT, Tunisia</i> <i>May 2026</i>
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- Built a discrete-event simulation model (ARENA) tracking member flows across 5 fitness center areas.
- Adjusted input data using ARENA Input Analyzer to determine distribution laws (Exponential, Triangular, Normal) for arrivals and service times.
- Executed 30 replications (960 min each), identifying weight-training as the bottleneck (95.6% utilization, up to 49 min wait).
- Expanded equipment capacity (10 → 12 machines), cutting wait times below 3 minutes and dropping utilization to 83.4%.

MLOps Pipeline for Predictive Maintenance of Turbofan Engines

Data Mining Project

ENIT, Tunisia

Jan 2026 – Mar 2026

- Developed an end-to-end pipeline for predicting the remaining useful life (RUL) of turbofan engines: C-MAPSS (NASA) data ingestion, preprocessing, deep learning model training, containerization, and automated deployment.
- Compared seven architectures (LSTM, CNN-1D, BiLSTM, Transformer, hybrid models) using PyTorch, with the Transformer achieving the best performance (RMSE 20.28).
- Technologies: Python, PyTorch, scikit-learn, FastAPI, Streamlit, Docker, Nginx, Hugging Face Spaces.

Simulation Model for Joint Quality-Maintenance Optimization

Year-End Project 2

ENIT, Tunisia

Nov 2025 – Apr 2026

- Conducted a literature review on maintenance strategies, quality control tools (SPC, Six Sigma, AMDEC), and existing analytical and simulation-based approaches for integrating quality and maintenance decisions in production systems.
- Designed and implemented a discrete-event simulation model under ARENA, comparing degradation-based, quality-based, and coupled maintenance strategies through custom decision logic, cost tracking, and key performance indicators.
- Performed comparative analysis and validated the model against a reference analytical study, demonstrating that the coupled strategy outperformed separate approaches in terms of cost, product conformity, and machine availability, supporting data-driven maintenance decision-making.

Secure Multi-Site Network Architecture for Industrial Connectivity

Internship

TELECOM, Tunisia

Jul 2025 – Aug 2025

- Applied a structured project methodology (requirements capture, system architecture design, sequential planning) to design a secure multi-site infrastructure interconnecting geographically distributed sites, a system-thinking approach transferable to multi-plant industrial network integration.
- Modeled and validated a resilient system architecture (VPN tunneling, firewalls, redundancy considerations) ensuring data confidentiality, integrity, and service continuity — principles directly applicable to reliability and business continuity in industrial operations.
- Conducted systematic testing and validation (connectivity checks, traffic analysis) to verify system performance against defined requirements, reflecting a quality-driven, continuous-improvement approach relevant to Industry 4.0 and OT/IT network integration.

Oil Pipeline Network Optimization

Operations Research Project

ENIT, Tunisia

2025

- Mathematical modeling (MILP) and exact resolution using a solver (Python / PuLP).
- Optimal design of a crude oil pipeline network across 5 zones, selecting pipeline type and routing to satisfy 20 simultaneous demands at minimum installation cost.
- Development and benchmarking of two heuristic methods against the exact optimum (GAP analysis).

The Future of Banking: Innovation and Challenges

Bibliographic Project

ENIT, Tunisia

Nov. 2024 – Apr. 2025

- Conducted a bibliographic study on the digital transformation of the banking sector, examining innovations such as artificial intelligence, blockchain, cryptocurrencies, and online banking services.
- Evaluated key industry challenges including cybersecurity threats, competition from FinTech companies, and evolving regulatory requirements.

OTHER RELEVANT ACTIVITIES

Student Member

IEEE Student Branch, National Engineering School of Tunis (ENIT)

Oct 2024 – Present

- Participated in national IEEE flagships (e.g., TSYP - Tunisian Students and Young Professionals Congress), networking with engineering leaders and engaging in technical tracks.
- Contributed to the organization of local technical workshops and national IEEE events, enhancing student engagement in emerging technologies.
- Represented the Student Branch in national IEEE congresses and challenges, strengthening professional connections within the Tunisian IEEE network.

Senior Member

november 2024 - Present

Industrial Engineering Club, National Engineering School of Tunis (ENIT)

- Attended industry conferences and panels (e.g., "Beyond Technical Skills: The Competencies Shaping Tomorrow's Engineer," ENIT, Jan 2026) to strengthen professional network and industry insight.
- Contributed to organizing TIES V5.0 - "The Role of Technology in Sustainable Development" and TIES V6.0 - "Shaping the Future of Industry: Orchestrating Next-Gen Operations with Data and Generative AI," major conferences of Tunisian Industrial Engineers Solutions.
- Participated in company visits to gain firsthand exposure to industrial operations and professional environments.

Senior Member

September 2018 - June 2019

Tunisian Red Crescent

- Assisted in organizing first-aid workshops and public health awareness campaigns.
- Participated in emergency relief operations, food drives, and medical supply distribution.
- Organized and supported local blood donation drives, assisting with donor registration and awareness.

LANGUAGES

- **Arabic** – Native language
- **English** – B2 level
- **French** – B2 level